



Micro-welding of sapphire and metal by femtosecond laser

Rui Pan ^a  , Dong Yang ^a, Taoshuai Zhou ^a, Yinghao Feng ^a, Zhisen Dong ^a, Zhaoyang Yan ^a, Peng Li ^{b c}, Jin Yang ^d, Shujun Chen ^a  

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Abstract

A direct joining of sapphire and Fe–36Ni alloy was successfully realized via femtosecond laser micro-welding for the first time. A sound joint without any voids or microcracks was obtained with a narrow interface width less than 1 μm. There was no obvious element diffusion or metallurgical reactions at the interface. Sapphire and Fe–36Ni alloy were found chemically bonded and mechanically interlocked evidenced by jagged feature at the interface due to “cold” machining of femtosecond laser ablation. The highly localized femtosecond laser irradiation and smaller heat-affected zone contributed to the shear strength of the joint as high as 108.35 MPa. A higher laser scanning speed corresponded with less jagged feature and thermal stress due to the reduced thermal deposition at the interface. Proper micro-welding parameters were obtained for sapphire/Fe–36Ni alloy, and was verified in the direct joining of sapphire/steel and sapphire/silicon. It appears the femtosecond laser micro-welding technique is promising for direct joining of materials with large physical property disparities, and beneficial for manufacturing of optomechanical components at high precision, efficiency and performance.



Keywords

Femtosecond laser; Direct joining; Sapphire; Fe–36Ni alloy; Microstructure; Shear strength

1. Introduction

Sapphire is a transparent ceramic in the form of single crystal alpha-alumina. Because of its excellent chemical, thermal, optical, mechanical and dielectric properties, sapphire has been widely used in structures, such as optical windows, high precision instruments, high temperature pressure sensor substrate, LED substrates, smart phone camera lens covers [\[1\]](#), [\[2\]](#), [\[3\]](#), [\[4\]](#). Such optomechanical components usually need sapphire to be joined with a structural body or other functional parts [\[5\]](#), [\[6\]](#), [\[7\]](#), which could be a problematic step due to the chemical and physical property differences between the sapphire-based assemblies, like sapphire with metals [\[8\]](#), [\[9\]](#), [\[10\]](#). For example, Fe–36Ni alloy (also known as Invar 36 alloy) is one of the materials that may need to be assembled with sapphire for the applications of precision instruments/tooling, optical mounting, etc., since Fe–36Ni alloy exhibits uniquely low coefficient of thermal expansion (at 20–100°C of about $1.2 \times 10^{-6} \text{K}^{-1}$) [\[11,12\]](#).

Considering the extreme brittleness of ceramics, diffusion bonding [\[10,13\]](#) and brazing [\[14,15\]](#) are the most popular joining methods for sapphire. These two methods essentially need to adopt some form of bonding interlayers or filler metals to either promote the interfacial reaction between the assembly due to the chemical and thermal stability of ceramics or to tune the residual stress formed in the joint due to the large difference of thermal expansion coefficient between ceramic and metal [\[16,17\]](#). However, the introduction of interlayers or filler metals usually generate new issues, i.e., a reduced high-temperature resistance of the joint due to the melting point of the adopted interlayer, an increased residual stress level caused by the new formed phases at the interface. In addition, diffusion bonding and brazing need to experience heating/temperature-hold/cooling processes in an atmosphere-controlled furnace, having complex joining procedures and low joining efficiency. Considering these limitations, an alternative, direct joining technique without any form of bonding interlayer is highly expected.

Unfortunately, traditional direct joining methods like fusion welding are not applicable for ceramic bonding due to the wider heat-affect zone (HAZ) formed by the large thermal deposition of the heat source at the interface, which unsurprisingly leads to the fracture of ceramics. It seems the smaller the volume of interface material, the lower the total amount of thermal stress [16]. Hence, a new direct bonding technology with highly focused heat source and small HAZ might be a solution.

In recent years, ultrafast laser with high peak power (typically in the order of MW of peak power and GWcm^{-2} in intensity), ultra-short pulse duration (10^{-12} to 10^{-15} s) and small focal beam diameter (from a few to tens of micrometers) have attracted great interests [18,19] and is proved to be a new powerful tool for transparent material joining [[20], [21], [22]]. In 2005, Tamaki et al. [23] first proposed ultrafast laser welding method for transparent glasses by focusing the laser light at the interfaces of the two substrates to stimulate the nonlinear absorption process of ultrafast laser. Basically, the highly focused ultrafast laser beam results in a localized melting of the glasses at the interface and a smaller HAZ because of the short pulse duration, which could realize the direct joining of glass materials. In the later years, ultrafast laser welding has been mainly attempted to join different glass assemblies [[24], [25], [26], [27]]. In 2006, Watanabe et al. [24] realized the direct joining of borosilicate glass and fused silica with a joint strength of ~ 16 MPa by an 800 nm, 85 fs and 1 kHz fs laser. This was the first report of ultrafast laser welding for dissimilar transparent materials. By improving and optimizing the energy and the repetition frequency of the applied ultrafast laser, the joint strength of dissimilar transparent glasses could be improved to over 100 MPa [[25], [26], [27]]. Usually, the high joint strength was obtained with the surface to be joined in optical contact, which have a high requirement of surface roughness and flatness [[28], [29], [30]]. Therefore, in recent years, non-optical contact ultrafast laser welding has become a new research hotspot [[31], [32], [33]]. However, the reported joint strength of non-optical contact surfaces is usually lower than that of the optical contact ones [[28], [29], [30], [31], [32], [33]]. Jia et al. [33] proposed a multi-scan picosecond laser welding method to join the non-optical soda lime glass with a gap up to $\sim 5 \mu\text{m}$. But the strength of the obtained joints was only up to 6.5 MPa.

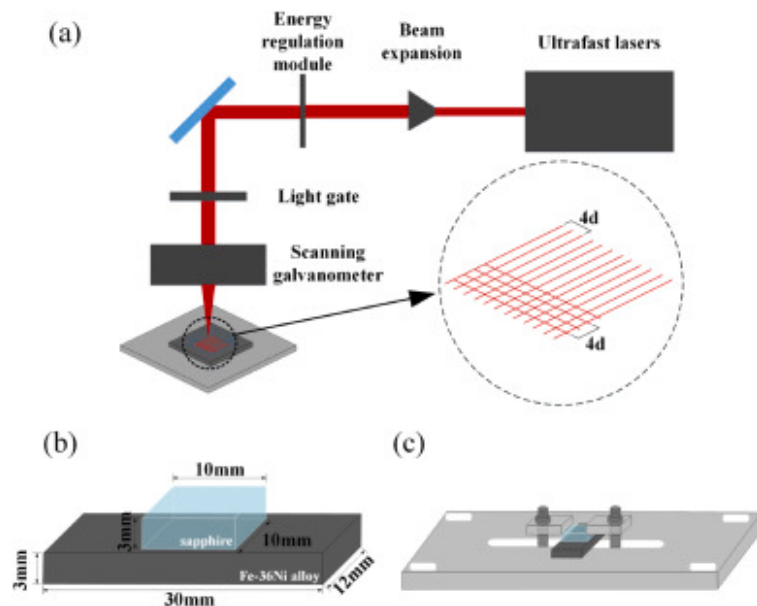
In 2019, Penilla et al. [34] extended the application of ultrafast laser welding to transparent alumina/alumina and zirconia/zirconia ceramic assemblies. However, probably due to the thermal stress sensitivity of ceramics and the large difference in thermal expansion coefficients of ceramic and metals, ultrafast laser based micro-welding has rarely been reported for the direct joining of sapphire and metals. The corresponding microstructures, mechanical property and joining parameters have not yet been revealed.

In the present study, we attempt to join sapphire/Fe–36Ni alloy assembly directly without any interlayer via femtosecond laser. By carefully tuning the laser energy deposition at the interface, sound sapphire/Fe–36Ni alloy joints with unique microstructures

and high shear strength were achieved. In addition, we investigated the effect of laser scanning speed on the microstructures and shear strength of sapphire/Fe-36Ni joints. We hope, the proposed femtosecond laser micro-welding method would provide a new strategy for the direct joining of the dissimilar materials with large difference in coefficient of thermal expansion.

2. Materials and methods

The schematic diagram of the equipment setup for femtosecond laser welding used in the present study is shown in Fig. 1(a). The femtosecond laser originates from Light Conversion emitting 280fs~2ps pulses with 1030nm wavelength, 1 kHz~1 MHz pulse repetition rate and 40W maximum average power. The commercially available sapphire samples with a size of 10mm×10mm×3mm used in this study are double-side polished. The Fe-36Ni alloy was cut into 12mm×30mm×3mm samples by electrical discharge machining. The surface to be joined of Fe-36Ni alloy samples were ground by sandpapers with different grades and subsequently polished with nylon polishing cloth. Then, the sapphire and the polished Fe-36Ni alloy samples were ultrasonically cleaned in absolute ethanol for 10min and then blow dried.



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Fig. 1. (a) Schematic of the equipment for femtosecond laser micro-welding of sapphire/metal; (b) The assemble of sapphire and Fe–36Ni alloy samples; (c) The fixture for a tight physical contact of the sample surfaces to be joined.

Prior to joining, the sapphire and Fe–36Ni alloy samples were assembled as shown in Fig. 1(b) and the surfaces to be joined were kept into optical contact by a fixture as shown in Fig. 1(c) to avoid the escape of the generated plasma from the interface gap during femtosecond laser irradiation. During joining, the focused laser beam scans vertically at the interface of sapphire/Fe–36Ni at 200kHz with scan speeds ranging from 50–250mm/s and an output power of 16W, following a pattern of cross lines using x-y galvanometric scanner and f - θ lens. The scan speeds are correlated with energy density based on Eq. (1) [35].

$$E_a = \frac{Df}{v} \times \frac{4E}{\pi d^2} \quad (1)$$

where E_a is the energy density, E is the single pulse energy and D is the focused laser spot diameter, D is the spot diameter, v is the laser scanning speed and f is the laser repetition frequency.

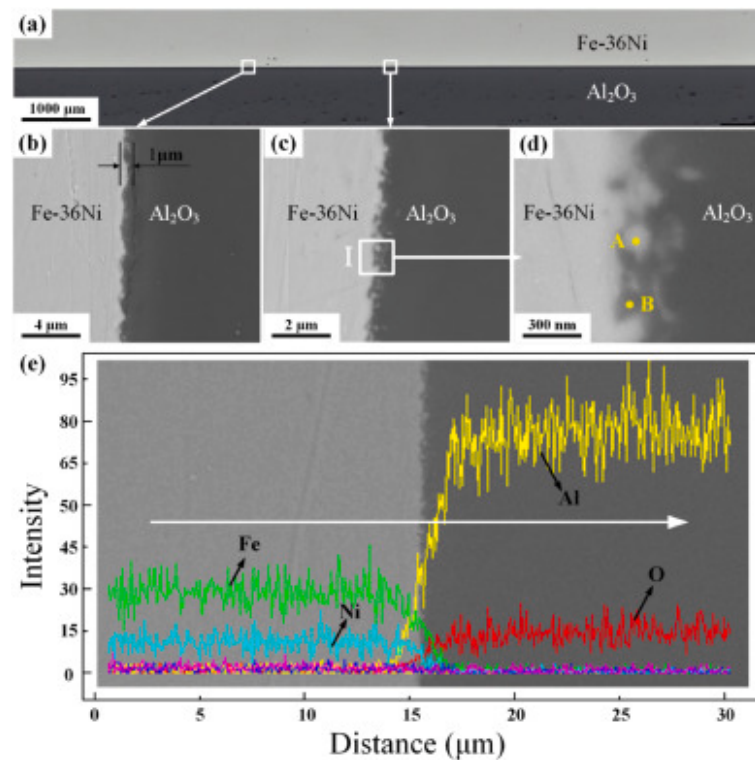
After joining, the cross sections of different sapphire/Fe–36Ni joints were cut via diamond wire cutting machine and polished for further analysis by scanning electron microscopy (SEM, Quanta 650) equipped with an energy dispersive spectroscopy (EDS, AMETEK), and also by the transmission electron microscopy (TEM, JEM-2100). The macroscale morphology and the 3D morphology of the sapphire/Fe–Ni alloy joining interface were analyzed by laser scanning confocal microscope (LSCM, Olympus OLS4100). Mechanical performance has been characterized by shear strength test. The testing was performed with a universal mechanical testing machine (MTS-300) equipped with a fixture enabling the pressing forces to be imposed with a constant displacement rate of 0.5mm/min. At least three specimens were tested to determine the average strength value of each experimental condition. After shear test, the fracture surface morphologies at both Fe–36Ni alloy and sapphire side were analyzed by SEM.

3. Results and discussion

3.1. Microstructures

The typical microstructure of the resultant sapphire/Fe–36Ni joints welded by femtosecond laser is shown in Fig. 2(a)~(d). It can be observed that sapphire and Fe–36Ni substrates are well joined by femtosecond laser and no voids or microcracks are formed

at the interface. The joint domain is composed of a 1 μm wide zone with white nano-scale particles and black matrix, as shown in Fig. 2(b). Based on the EDS analysis in Table 1, it is observed that the white nano-scale particle marked in microzone A is Fe–36Ni alloy and the black matrix microzone B is sapphire in Fig. 2(d). From the line scan result across the interface and the composition mapping of the selected zone in Fig. 2(e), it is observed that there is no obvious element diffusion or reaction at the interface, indicating a sound chemical bond of interfacial materials, which could greatly reduce the residual stress caused by the lattice mismatch due to the element diffusion and the formation of new phases. In addition, as can be seen in Fig. 2(b)~(d), the interface shows a morphologically jagged feature due to the scanning of the focused femtosecond laser beam at the interface, indicating that the laser focal volume actually modifies the surfaces of both sapphire and metal.



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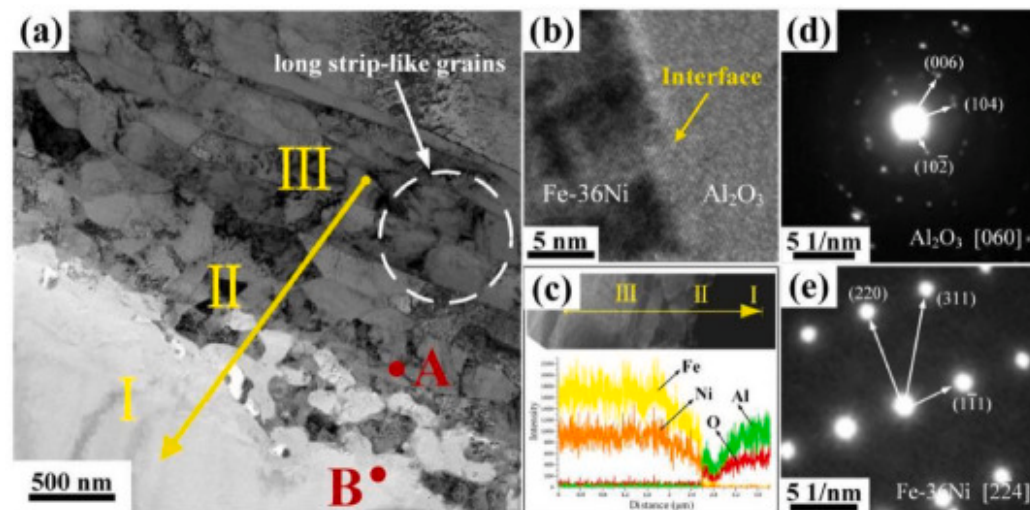
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Fig. 2. (a)~(d) The typical microstructure of sapphire/Fe–36Ni joint welded by femtosecond laser; (e) EDS line scan result across the interface along the white line marked in the figure.

Table 1. EDS compositions of the micro-zones marked in joint shown in Fig. 2(d).

Element (at.%)	Al	O	Fe	Ni	Possible Phases
Region A	1.27	1.43	62.50	34.80	Fe–36Ni
Region B	41.19	58.37	0.4	0.04	Al ₂ O ₃

In order to have a better understanding of phases and bonding features of the joint domain formed by femtosecond laser based micro-welding, TEM analysis was performed for the selected zone I marked in Fig. 2(c), as shown in Fig. 3. The overall morphology of the selected zone fabricated by focused ion beam (FIB) is shown in Fig. 3(a). It can be seen from the line scan in Fig. 3(c) that the zone I is the sapphire, zone II is the interface domain and zone III is the Fe–36Ni alloy. Zone II is composed of phases A and B, which is proved to be Fe–36Ni alloy particles and sapphire particles according to the selected area electron diffraction (SAED) patterns presented in Fig. 3(e) and (d). The high-resolution transmission electron microscopy (HRTEM) images of a Fe–36Ni particle and sapphire phase at the interface are shown in Fig. 3(b). It can be clearly seen that a bonding interface is formed due to the chemical bonding between Fe–36Ni and sapphire phases, verifying that there are no other phases formed in between of them. This might be attributed to two reasons. One is the metallurgical inactivity between Fe–36Ni and sapphire, and the other is the high localization and the small thermal effect of femtosecond laser irradiation, which cannot trigger the interface reaction between Fe–36Ni and sapphire.



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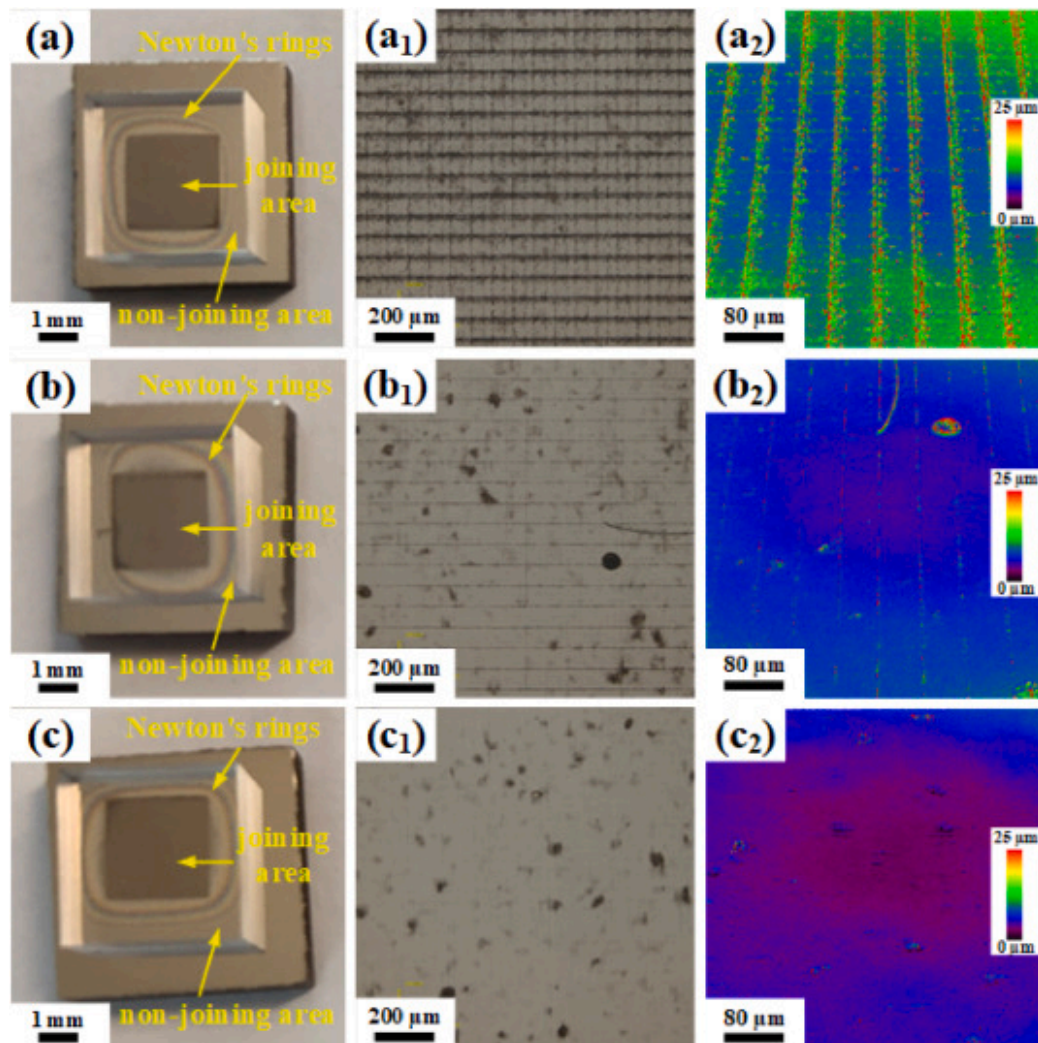
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Fig. 3. TEM analysis at the interface of Fe–36Ni/sapphire joint: (a) Bright field image of sapphire/Fe–36Ni joint domain; (b) The HRTEM images of Fe–36Ni/sapphire interface; (c) The line scan result along the yellow line marked in (a); (d)~(e) The SAED patterns of the different phases. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

In addition, it can be observed in Fig. 3(a) that the grains of the Fe–36Ni alloy at the place a bit away from the interface are larger and in a shape of longer strip than those of the grains around the interface. It is inferred that the long strip-like grains are the base metal grains and the small particles at the interface are resulted from the irradiation of the femtosecond laser beam. When the femtosecond laser beam focused at the interface of the two substrates, ionization is initiated by nonlinear/linear absorption of the base materials in the focal volume, resulting in the creation of plasma consisting of laser-induced energetic species such as atoms, clusters, droplets, and particulates. The generated plasma of the upper and lower material in the focal volume are mixed together by their expansion along the interface gap within a localized zone, thus resulting in a mixed phase zone after the solidification of the modified area by femtosecond laser. The mixed phases around the interface also demonstrate that the modification of femtosecond laser occurs in both of the two substrates around the interface, which can also be proved by the fracture surface morphology of the joint in the following text.

3.2. Effect of laser scanning speed

The results of the control experiment are shown below, that the macroscopic morphologies of sapphire/Fe–Ni alloy assemblies, joined by femtosecond laser with different laser scanning speeds at 200kHz repetition frequency and 16W output power when the line scan space is 80μm, are shown in Fig. 4. As can be seen in Fig. 4(a), (b) and (c), all the assemblies are well joined without any macro cracks or deformation. The domain exposed to the focused laser at the interface becomes darkened due to ablation of femtosecond laser. With the increasing of the laser scan speed from 50mm/s to 250mm/s (i.e., the energy density reduced from $1.02 \times 10^3 \text{ J/cm}^2$ to $2.04 \times 10^2 \text{ J/cm}^2$), the resultant laser-inscribed mark becomes less obvious as shown in Fig. 4(a₁), (b₁) and (c₁) and the corresponding 3D morphology of the joined interface via laser scanning get less rough and fluctuated as shown in Fig. 4(a₂), (b₂) and (c₂). It is because the number of equivalent pulses is decreasing as the scan speed increases, resulting in less laser ablation at the sapphire/Fe–36Ni alloy interface due to the less laser energy input. In addition, Newton's rings, which consists of colored fringes, can be observed in the non-joining area (besides to the joining area) on the assemblies, as shown in Fig. 4(a), (b) and (c). Newton's rings is a phenomenon in which an interference pattern is created by the reflection of light between two surfaces when the gap between two the surfaces varies from zero at the point of contact to a finite value in the wedge-shaped region [36]. Newton's rings are usually composed of a series of concentric circular. In present cases, Newton's rings are formed since the gap between sapphire and Fe–Ni alloy in the joining area is decreased to zero, while, the gap in the non-joining area is decreased to a value small enough to form Newton's rings.



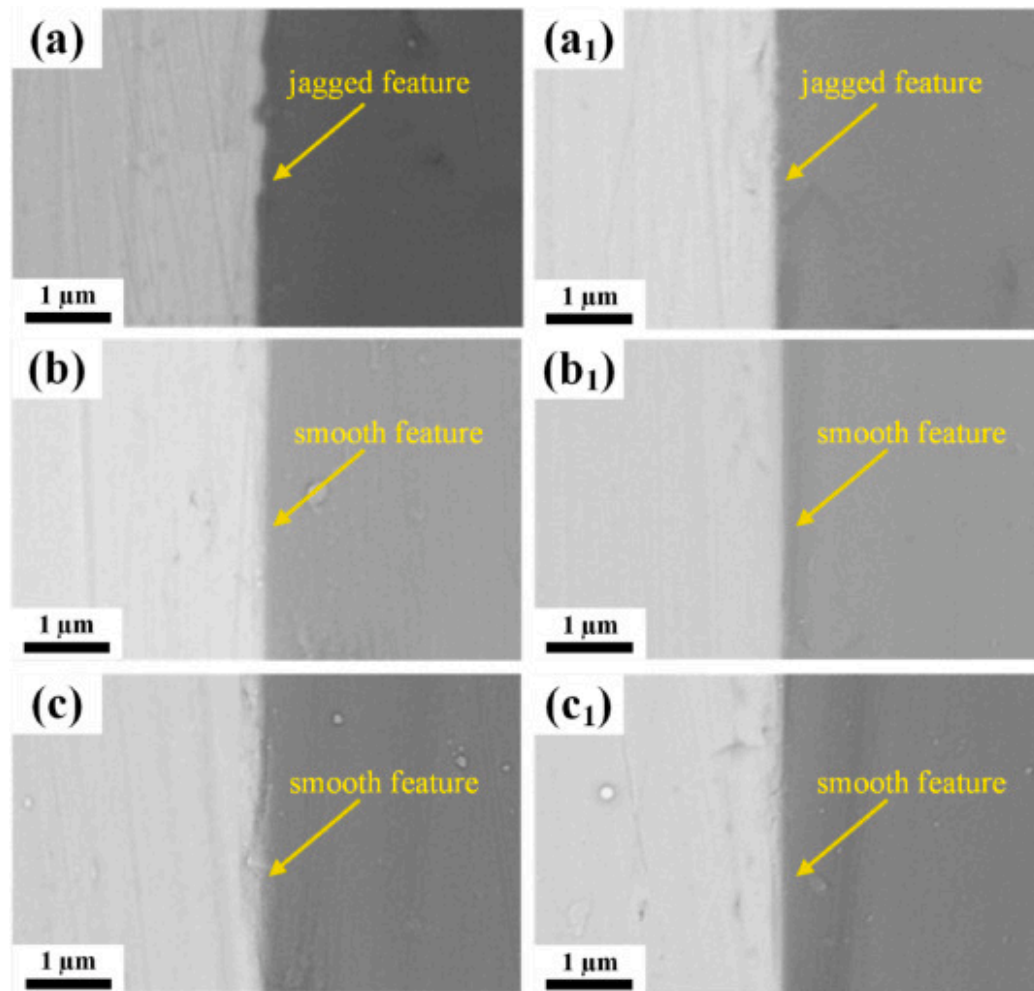
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Fig. 4. The macroscopic morphologies of sapphire/Fe–Ni alloy assemblies at different laser scanning speeds: (a)~(a₂) 50mm/s; (b)~(b₂) 100mm/s; (c)~(c₂) 200mm/s (a₂) ~ (c₂) are the LSCM image of the 3D morphology of the joined interface.

Correspondingly, the microstructures of the cross-sections of sapphire/Fe–36Ni interfaces joined by femtosecond laser at the different scanning speeds are shown in Fig. 5. It can be seen in Fig. 5(a) and (a₁) that the microstructures of the sapphire/Fe–36Ni

joints show morphologically jagged feature when the scanning speed is 50mm/s. As scanning speed increases to 100mm/s and 200mm/s, it can be seen in Fig. 5(b) and (b₁) and Fig. 5(c) and (c₁) that the microstructure of the sapphire/Fe-36Ni joints become smooth and the morphologically jagged feature disappears. It is corresponded to the 3D weld morphology at the interface, that the rugged 3D weld morphology matches the microstructure of morphologically jagged feature. It is because more mixed plasma will be generated at the joint interface due to the stronger ablation effect when the scanning speed is small, and the large amount of mixed plasma eruption and expansion is the main reason for the formation of morphologically jagged feature.



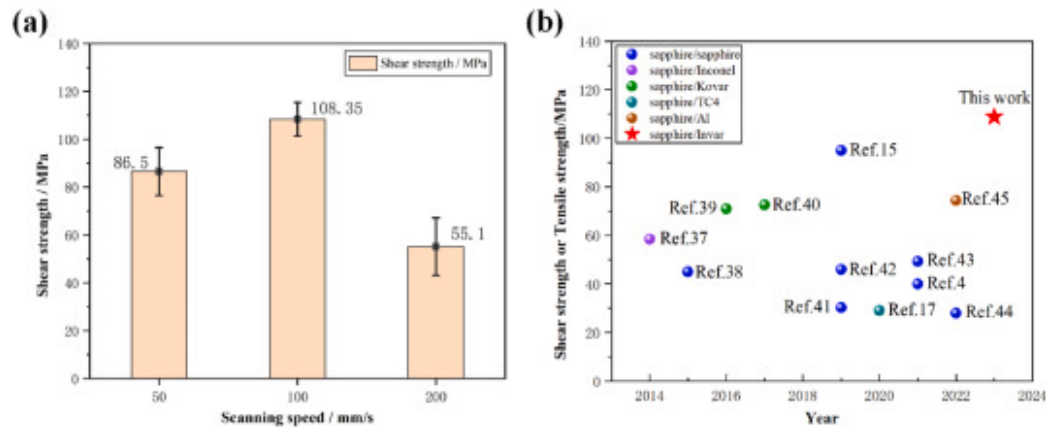
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Fig. 5. The microstructures of the cross-sections of sapphire/Fe–36Ni interfaces at different scanning speeds: (a)~(a₁) 50mm/s; (b)~(b₁) 100mm/s; (c)~(c₁) 200mm/s.

3.3. Mechanical property

Shear strength test were performed to characterize the mechanical property of sapphire/Fe–36Ni joints welded by femtosecond laser. Fig. 6(a) shows the shear strength of the sapphire/Fe–36Ni joints welded at different laser scanning speed. It can be seen that the shear strength first increases from 86.5MPa to the maximum value 108.35MPa with the scanning speed increasing from 50mm/s to 100mm/s and then decreases to 55.1MPa when the scanning speed further increases to 200mm/s. We compared the tested value of 108.35MPa with several typical reports offering data on shear strength or tensile strength of sapphire joints [4,15,17,[37], [38], [39], [40], [41], [42], [43], [44], [45]], as shown in Fig. 6(b) (More detailed information is shown in Table 2). It can be seen that our ultrashort-laser-welded sapphire/metal joints possess the highest shear strength among all the published reports, exhibiting an excellent mechanical property. More specifically, the shear strength of the sapphire/Fe–Ni alloy joints welded by femtosecond laser in this work is 50%–88% higher than the brazed sapphire/Fe–Ni alloy joints reported in Refs. [37,39] (which are 72.6MPa for sapphire/Kovar alloy assembly and 58.5MPa for Sapphire/Inconel assembly, respectively). Such an excellent mechanical property can be attributed to the high localization and the extremely small thermal effect of femtosecond laser irradiation, which could greatly reduce the thermal residual stress in the joint and enable the direct joining of two substrates with highly different coefficient of thermal expansion. In addition, there is no need for the femtosecond laser based micro-welding to experience heating and cooling processes in a furnace for the whole assembly, which not only improves the joining efficiency but also reduces the residual stress during the cooling process. This seems to be a big step forward, comparing with the tradition ceramic bonding methods like diffusion bonding and brazing.



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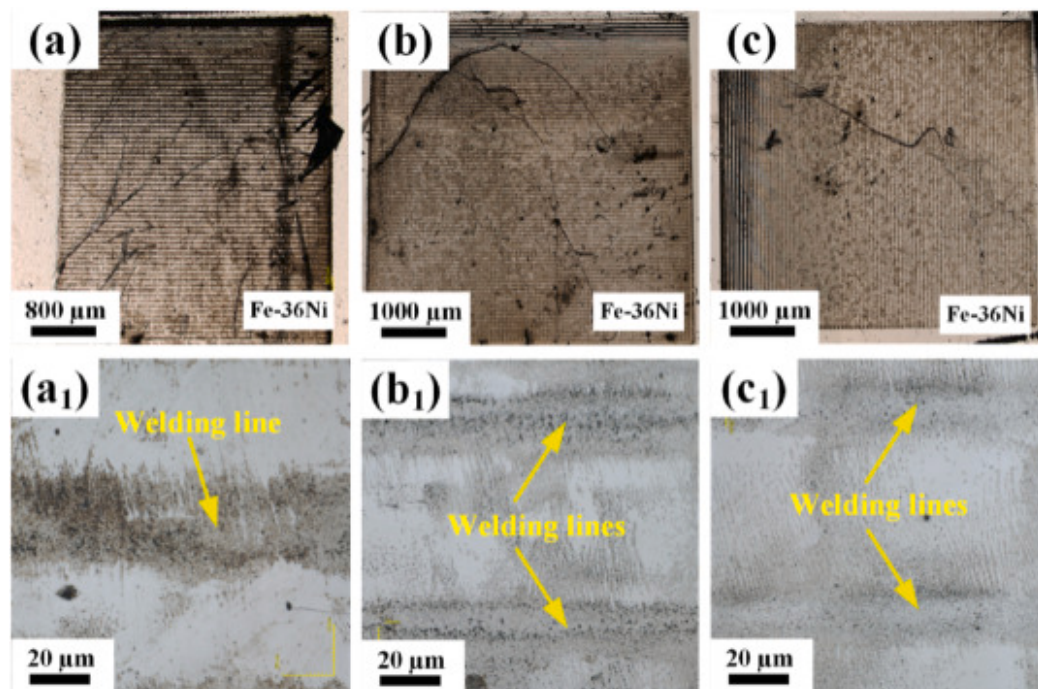
Fig. 6. (a) The shear strength of the resultant sapphire/Fe-36Ni joints at different scanning speeds; (b) The comparison of joint strength between the prepared sapphire/Fe-36Ni joints with several other representative reports.

Table 2. The detailed information of several reported sapphire joints.

Author	Year	Materials	Method	Strength type	Strength value	Ref.
T. Zaharinie et al.	2014	Sapphire/Inconel 600	Brazing	shear strength	58.5 MPa	[37]
W. Cui et al.	2015	Sapphire/Sapphire	Ultrasonic soldering	shear strength	43–48 MPa	[38]
C. Xin et al.	2016	Sapphire/Kovar	Brazing	tensile strength	69.6–73.1 MPa	[39]
C. Xin et al.	2017	Sapphire/Kovar	Brazing	tensile strength	72.6 ± 4.4 MPa	[40]
D. Mu et al.	2019	Sapphire/Sapphire	Brazing	shear strength	30.3 MPa	[41]
Y. Xu et al.	2019	Sapphire/Sapphire	Ultrasonic soldering	shear strength	46 MPa	[42]
W. Guo et al.	2019	Sapphire/Sapphire	Brazing	shear strength	95 MPa	[15]
Z. Fan et al.	2020	Sapphire/TC4	Brazing	shear strength	29.1 MPa	[17]

Author	Year	Materials	Method	Strength type	Strength value	Ref.
Y. Xu et al.	2021	Sapphire/Sapphire	Ultrasonic soldering	shear strength	40MPa	[4]
W. Guo et al.	2021	Sapphire/Sapphire	Brazing	shear strength	49.3MPa	[43]
X. Ma et al.	2022	Sapphire/Sapphire	Ultrasonic soldering	shear strength	28MPa	[44]
J. Lv et al.	2022	Sapphire/Al	Ultrasonic soldered	shear strength	74.42MPa	[45]
R. Pan et al.	2023	Sapphire/Invar	Femtosecond laser	shear strength	108.35MPa	This work

The macro-scale fracture surface morphology of the sapphire/Fe–36Ni joints at different scanning speeds after shear strength tests are shown in Fig. 7. It can be seen in Fig. 7(a) that the fracture occurs partially at the interface when the scanning speed is 50mm/s, and a few sapphire debris are attached on the Fe–36Ni alloy fracture surface, indicating some residual stress is generated at sapphire side near the interface. When the laser scanning speed raise to 100mm/s and 200mm/s, as can be seen in Fig. 7(b) and (c), the fracture occurs almost entirely at sapphire/Fe–36Ni interface, indicating a low residual stress level generated at sapphire side near the interface. The magnified images of Fig. 7(a), (b) and (c) are shown in Fig. 7(a₁), (b₁) and (c₁), it can be seen that the laser induction and ablation marks at the interface gradually weaken as the scanning speed increases due to the reducing accumulated laser energy input.



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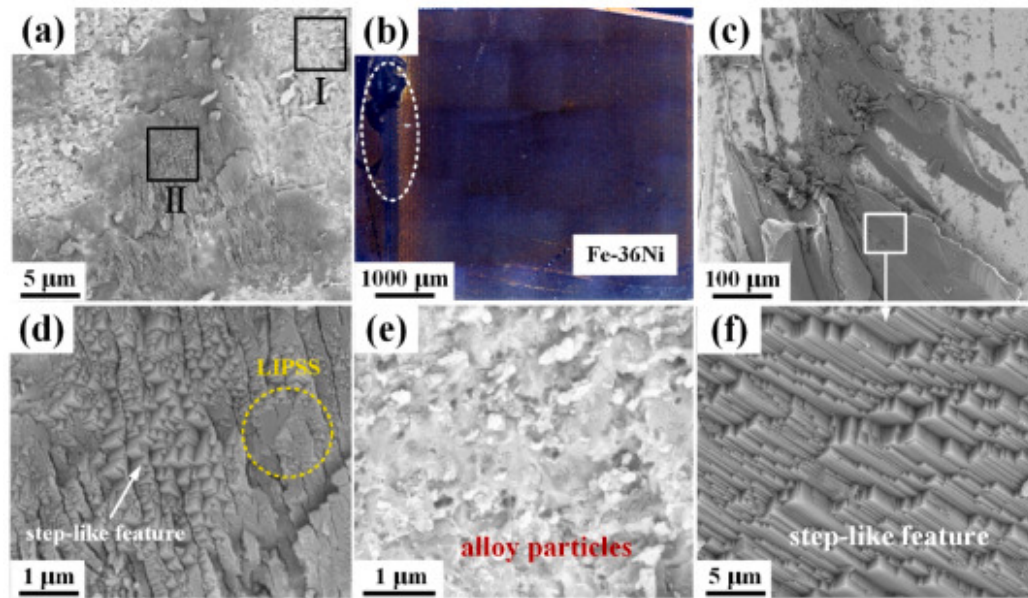
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Fig. 7. The fracture surface morphology of the sapphire/Fe–36Ni joints at different scanning speeds: (a)~(a₁) 50mm/s; (b)~(b₁) 100mm/s; (c)~(c₁) 200mm/s.

According to the fracture surface morphology analysis, the reason for the change in the shear strength of the resultant sapphire/Fe–36Ni joints is known that when the scanning speed is too slow, less than 100mm/s, the larger number of equivalent pulses and stronger thermal input may result in larger thermal accumulation and stress at the interface, leading to relatively lower shear strengths. However, if the scanning speed is too fast, although the formation of thermal stress is correspondingly reduced, the induction and ablation of the laser beam also get weaker, that the generated mixed plasma necessary for the interface joining effect also get diminished, thus leading to a decrease of the shear strength overall.

To analyze the typical micro-scale fracture surface morphology of the joints, sapphire/Fe–36Ni alloy joints welded at a laser speed of 50mm/s were selected as a presentation. The typical micro-scale fracture surface morphology at Fe–36Ni alloy and

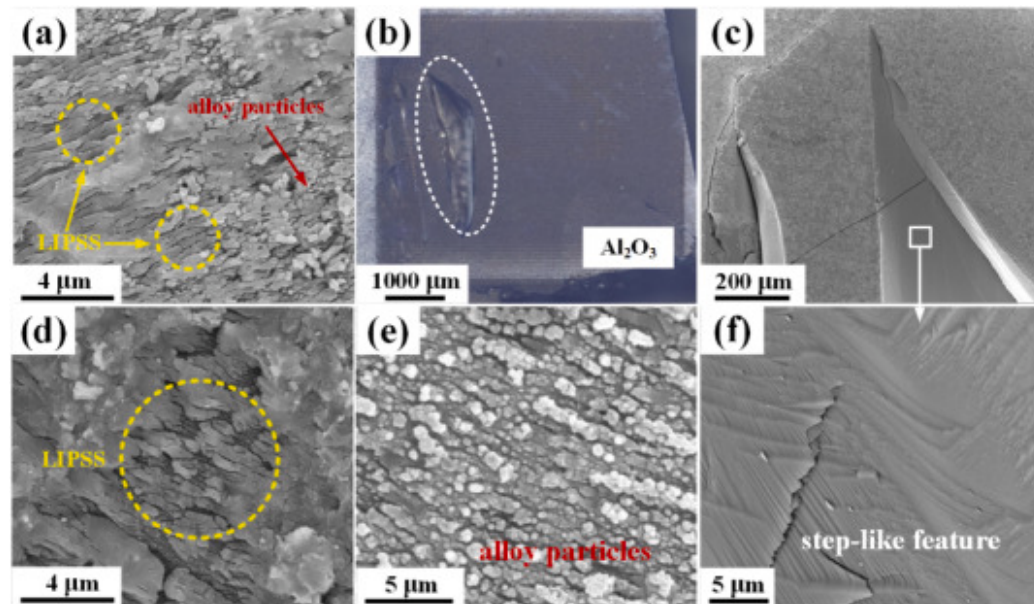
sapphire side after the shear strength test are shown in Fig. 8, Fig. 9, respectively. As can be seen in Fig. 8(b), the fracture occurs mainly at the interface of sapphire/Fe–36Ni alloy joint, and only one small path penetrates the sapphire substrate. This is further verified by the morphology of the fracture surface at sapphire side in Fig. 9(b). The morphology of the interfacial fracture zone is shown in Fig. 8(a), (d) and (e). It can be observed that the interfacial zone presents dark zone II and white zone I in Fig. 8(a), which can be identified to be sapphire and Fe–36Ni alloy, respectively, by the EDS analysis in Table 3. The co-existence of sapphire and Fe–36Ni alloy in the interfacial fracture surface is due to the morphologically jagged feature of the joint domain, which may provide a mechanical interlocking effect during shear strength test and thus contributes to the joint strength.



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Fig. 8. The typical fracture surface morphologies of Fe–36Ni/sapphire joints (welded with a laser scanning speed of 50mm/s) after shear strength tests at Fe–36Ni alloy side: (a) the microstructure of the fracture surface occurred at the interface; (b) the macroscopic morphology of the fracture surface at Fe–36Ni alloy side; (c) the microstructure of the fracture surface located in the sapphire substrate close to the interface; (d) the magnified image of the marked zone II in (a); (e) the magnified image of the marked zone I in (a); (f) the magnified image of the marked zone in (c).



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Fig. 9. The typical fracture surface morphology of Fe–36Ni/sapphire joints (welded with a laser scanning speed of 50mm/s) after shear strength tests at sapphire side: (a) the microstructure of the fracture surface occurred at the interface; (b) the macroscopic morphology of the fracture surface at sapphire side; (c) the microstructure of the fracture surface occurred in sapphire; (d) the LIPSS formed at fracture surface occurred at the interface; (e) the alloy particles attached at fracture surface occurred at the interface; (f) the magnified image of the marked zone in (c).

Table 3. EDS compositions of the micro-zones marked shown in Fig. 8(a).

Element (at.%)	O	Al	Fe	Ni	Possible phases
Region I	1.98	4.84	64.15	29.03	Fe–36Ni
Region II	58.17	30.92	7.3	3.61	Al ₂ O ₃

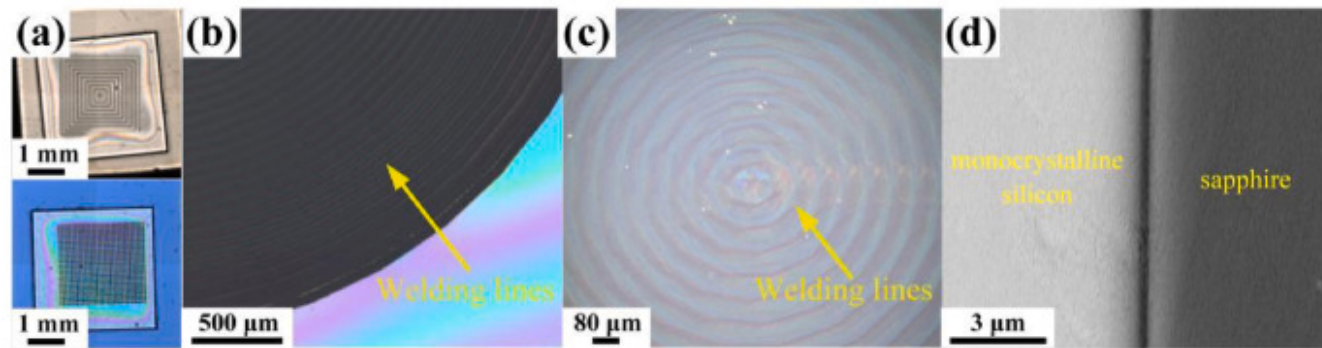
The sapphire domain exhibits a step-like morphology on different ripple-like structures, showing a feature of brittle fracture, as shown in Fig. 8(d). The ripple-like structure is a typical ultrafast laser-induced structure and is named as laser-induced periodic surface structure (LIPSS). The Fe–36Ni alloy domain is composed of alloy particles, as can be observed in Fig. 8(e). The formation of alloy particles is a result of laser ablation at the metal interface. Fig. 8(c) and (f) present the fracture surface morphology of the crack located in the sapphire substrate close to the interface. The step-like feature shows obvious brittle fracture, matching the hard-brittle characteristic of the sapphire substrate.

Fig. 9 is the morphology of the fracture surface at sapphire side, showing similar features compared to fracture surface at metal side. It can be observed that the interfacial fracture surface in Fig. 9(a), (d) and (e) show clearer features of ripple-like LIPSS in the laser-modified zone of sapphire at the interface compared to the corresponding surface at metal side. For a particular material, the LIPSS can only be formed by the laser irradiation when laser fluence is slightly above the ablation threshold, while the ablated structures like random micro/nano particles or periodical microcon structures are generated at a higher laser fluence [46]. Since the ablation threshold of sapphire is higher than that of metal, when LIPSS can be just generated at sapphire interface, the laser fluence is definitely higher than the ablation threshold of Fe–36Ni alloy and is able to ablate the metal interface. This is the reason that the fracture surface at sapphire side shows a more obvious LIPSS feature and the corresponding metal side exhibits an ablated structure feature. All these features contribute to the formation of a sound joint by femtosecond laser welding.

It has been widely reported that the surface structures formed by femtosecond laser treatment is determined by the amount of the femtosecond laser energy deposition at the surface [[47], [48], [49]]. A large amount of energy deposition can result in laser ablated micro/nano structures at the interface. As we stated in the manuscript that the joining of Fe–36Ni/sapphire is attributed not only to the chemical bonding but also the mechanical interlocking featured by the jagged morphology at the interface. A larger scale ablated microstructures may contribute more to the mechanical interlock effect than small scale structures, like LIPPs. However, the ablation microstructure also means a larger laser energy deposition at the interface and a larger thermal stress may be formed in sapphire, thus deteriorating the quality of the joint. Therefore, only with a reasonable interface structure feature determined by suitable parameter combinations, can a sound joint be obtained.

3.4. Verification of micro-welding of sapphire with other materials

Several other different materials' assemblies were also attempted to be joined by ultrafast laser micro-welding, as shown in Fig. 10. It can be seen that all the assemblies are well joined and the welded seams are nicely and uniformly formed. There are no microcracks generated at the interfaces. In addition, Newton's rings can also only be observed in the non-joining area on the assemblies (not in the joining area), as shown in Fig. 10 (a), indicating a sound joining of the area irradiated by femtosecond laser. All these features indicate the good versatility of ultrafast laser to realize the high precision, high efficiency and high-performance direct joining of different material combinations with large physical property disparities [[18], [19], [20], [21], [22], [23], [24]].



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Fig. 10. Other joints of sapphire and other material combinations welded by femtosecond laser: (a) The joints of sapphire/stainless steel (the upper figure) and sapphire/monocrystalline silicon (the lower figure); (b)~(c) The welding lines at sapphire/monocrystalline silicon interface; (d) The microstructure of a sapphire/monocrystalline silicon joint. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

4. Conclusion

In summary, the direct joining of sapphire/Fe–36Ni alloy joints has been successfully realized by femtosecond laser micro-welding. The interface of the sapphire/Fe–36Ni joints were well-joined without any voids or microcracks, and the joint domain consisted of a mixed phase region with Fe–36Ni particles and sapphire in a width less than 1 μm. The shear strength of sapphire/Fe–36Ni joints could reach 108.35 MPa, which was 50%–88% higher than that of the reported diffusion bonded or brazed

sapphire/metal joints. The bonding mechanism can be attributed to the chemical bonding between sapphire/Fe–36Ni and a mechanical interlocking effect due to the morphologically jagged feature at the interface, resulted from the “cold” machining feature of femtosecond laser ablation. Increasing the laser scanning speed results in less jagged feature and residual stress at interface of the joints due to the reduced thermal deposition at the interface. Our study demonstrates that femtosecond laser based micro-welding could be a convenient, flexible and feasible method for the direct joining of sapphire and metals with high precision, high strength and high efficiency.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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